Lemma. Let $G$ be a group and $A$ be a set.
For an Action of $G$ on $A$, $|G:G_a| = |Ca|$
Proof. Since for any $g_1, g_2 \in G_a$, $g_1 g_2^{-1} a = g_1 a = a$, so $g_1 g_2^{-1} \in G_a$.

Thus $G_a$ is a subgroup. And, define a map

$$C_a \to \{gG_a \mid g \in G\} \; ; \; g \cdot a \mapsto gG_a$$

Then, $g_1 G_a = g_2 G_a$ implies $g_1^{-1} g_2 \in G_a$, or $g_1^{-1} g_2 a = a$, or $g_1 a = g_2 a$.
So injective, and the surjective is trivial. Thus, Proved.

Thm. Let $G$ be a group and $H$ be a subgroup of $G$.
An action $G \times \{gH \mid g \in G\} \to \{gH \mid g \in G\} \; ; \; (g_1, g_2 H) \mapsto g_1 g_2 H$
satisfies:
1) It acts transitively. (It is clear, since $\{gH \mid g \in G\}$ is a partition of $G$).
2) $G_H = H$
3) $\ker \pi_H = \bigcap_{x \in G} xHx^{-1}$ is the largest normal subgroup of $G$ contained in $H$.

Proof. $\ker \pi_H = \{g \in G \mid gxH = xH \text{ for all } x \in G\}$
$= \{g \in G \mid x^{-1} g x H = H \text{ for all } x \in G\} = \bigcap_{x \in G} xHx^{-1}$.
Let $g \in G$ and $h \in \ker \pi_H$. Then, for any $x \in G$,
$x^{-1}(ghg^{-1})x = (g^{-1}x)^{-1} \cdot h \cdot (g^{-1} \cdot x) \in \ker \pi_H$, so $\ker \pi_H \trianglelefteq G$.

And, given $g \in \ker \pi_H$ and $h \in H$, $h^{-1} g h H = H \Rightarrow g \in H$, so $\ker \pi_H \leq H$.
Finally, if $N \leq H$ and $N \trianglelefteq G$, then given $x \in G$

$$N = xNx^{-1} \leq xHx^{-1}, \text{ so } N \leq \bigcap_{x \in G} xHx^{-1}$$

Corollary. If $G$ is a finite group of order $n$ and $p$ is the smallest prime dividing $n$, then any subgroup of index $p$ is normal.
Proof. Let $H \leq G$ such that $|G:H| = p$. And, let $K = \ker \pi_H$, $\pi_H$ is an action above.
Since $K \leq H$, let $|H:K| = k$. Then, $|G:K| = |G:H| \cdot |H:K| = p \cdot k$.
Since $H$ has $p$ left cosets, $G/K \cong \pi_H[G] \leq S_p$, so $pk = |G/K|$ divides $p!$,
so $k = 1$, $H = K$.

## Automorphism of Group.

**Prop.** Let $H$ be a normal subgroup of the group $G$.
Then, $G$ acts by conjugation on $H$ as automorphisms of $H$.
More specifically, for each $g \in G$,

$$H \to H : h \mapsto ghg^{-1}$$

is an automorphism of $H$. Moreover, the permutation represent,

$$T_H : G \to S_H : g \mapsto \sigma_g \quad \text{where } \sigma_g(h) = ghg^{-1}$$

is a Homomorphism of $G$ into $Aut(H) \leq S_H$, with $\ker T_H = C_G(H)$.
In summary, $G/\ker T_H = G/C_G(H) \cong \operatorname{Im} T_H \leq Aut(H)$, and $G/Z(G) \cong \operatorname{Inn}(G)$.

**Proof.** Since $H$ is normal, the Action is well-defined. And,

$$\ker T_H = \{g \in G \mid \sigma_g = id\} = \{g \in G \mid ghg^{-1} = h \text{ for all } h \in H\} = C_G(H).$$

**Corollary.** For any subgroup $H$ of $G$,

$N_G(H)/C_G(H)$ is isomorphic to a subgroup of $Aut(H)$.

**Prop.** $Aut(\mathbb{Z}_n) \cong \mathbb{Z}_n^\times$.

**Proof.** Denote $\mathbb{Z}_n = \langle x \rangle$, where $x \in \mathbb{Z}_n$. Given automorphism $\varphi \in Aut(\mathbb{Z}_n)$, the $\varphi$ is uniquely determined by $\varphi(x) = x^a$, the integer $1 \leq a \leq n$, and $|x| = |\varphi(x)| = |x^a|$, so $\gcd(a,n) = 1$. Now,

$$Aut(\mathbb{Z}_n) \to \mathbb{Z}_n^\times : \varphi_a \mapsto a \bmod n$$

is the isomorphism.

$$\mathbb{Z}_8^\times = \{1,3,5,7\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

**Note:** If $P$ is a group of a prime order $p$, then

$$Aut(P) \cong \mathbb{Z}_p^\times \cong \mathbb{Z}_{p-1}.$$

Lemma. Let $P \in Syl_p(G)$. If $Q$ is $p$-subgroup of $G$, then $Q \cap N_G(P) = Q \cap P$.

Proof. Since $P \subseteq N_G(P)$, $Q \cap P \subseteq Q \cap N_G(P)$ is trivial.

Since $Q \cap N_G(P) \leq N_G(P)$, denote $H = Q \cap N_G(P)$,

$$PH = \bigcup_{h \in H} Ph = \bigcup_{h \in H} hP = HP$$

Thus $PH \leq G$ and $|PH| = \frac{|P| \cdot |H|}{|P \cap H|}$. By the Lagrange's Thm, $H \leq Q$ means order of $H$ is power of $p$. Therefore, $P \leq PH$ and $|PH|$ is power of $p$, thus $PH = P$, or $H \leq P$. Now, $H = Q \cap N_G(P) \leq Q$, $H \leq Q \cap P$. ∎

Lemma. If $G$ is finite abelian group and $p$ is a prime number dividing $|G|$, then $G$ has an element of order of $p$.

Proof. Using induction for $|G|$. If $|G| = 1$, trivial.

Now, suppose that the statement hold for all groups whose order is smaller than $|G|$.

Since $|G| > 1$, take $x \in G$ with $x \neq 1$. That is, $|x| > 1$.

If $|G| = p$, then $|x| = p$, by the Lagrange's Theorem. ($|x| \mid p$).

Now, suppose $|G| \neq p$. ($|G| > p$).

If $p$ divides $|x|$, $|x| = pn$ for some $n \in \mathbb{N}$, so $|x^n| = p$. So clear.

If $p$ does not divides $|x|$, let $N = \langle x \rangle$. (Clearly, $p \nmid |N| = |x|$).

Since $G$ is abelian, $N$ is Normal, and $x \neq 1$, so it is non-trivial.

By the Lagrange Theorem, $|G| = |G/N| \cdot |N|$. But, $p$ does not divide $|N|$, $p$ divides $|G/N|$. Now, using the hypothesis of the induction, $G/N$ has an element of order $p$, denote this $\bar{y} = yN$. Since $\bar{y} \neq \bar{1}$, $y \notin N$. But, $\bar{y}^p = \bar{1}$, so $y^p \in N$. This implies $\langle y^p \rangle \subsetneq \langle y \rangle$. Since $|y^p|$ divides $|y| = p$ the only possibilities are $|y^p| = 1$, this $y$ is we desired.

Theorem.
Let $G$ be a group of order $p^a \cdot m$ where $p^a$ is a power of prime $p$, and $p \nmid m$.

1st sylow theorem. Sylow $p$-subgroup of $G$ exists.

2nd sylow theorem. If $P \in Syl_p(G)$ and $Q$ is $p$-subgroup of $G$, then $Q \leq gPg^{-1}$ for some $g \in G$.

3rd sylow theorem. $n_p \equiv 1 \mod p$ and $n_p = |G : N_G(P)|$ for any $P \in Syl_p(G)$.

Proof. To show the Existence of Sylow $p$-subgroup, using induction for $|G|$.
If $|G|=1$, then $p^a=1$, so $Syl_1(G) = \{G\} \neq \emptyset$.
Assume that the sylow $p$-subgroup always exists for any group of order smaller than $G$.
If order of $Z(G)$ is divided by $p$, then by the Cauchy's Theorem for abelian group, there exists order of $p$ element $y \in Z(G)$. Let $N = \langle y \rangle$. Denote $\overline{G} = G/N$.
By the hypothesis and the Lagrange's Theorem, $|\overline{G}| = |G|/|N| = p^{a-1} \cdot m$.
By the induction, $\overline{G}$ has a sylow $p$-subgroup $\overline{P}$ of order $p^{a-1}$.
By the correspondence isomorphism Theorem, there is $P \leq G$ such that $\overline{P} = P/N$.
By the Lagrange's Theorem again, $p^{a-1} = |\overline{P}| = |P/N| = |P|/p$, so $|P| = p^a$.
Therefore, there exists a sylow $p$-subgroup $P$, in case of $p \mid |Z(G)|$.
If order of $Z(G)$ is not divided by $p$, let $g_1, \cdots, g_r$ be representatives of distinct non-central conjugacy classes of $G$. The class equation gives:

$$|G| = |Z(G)| + \sum_{i=1}^{r} |G : C_G(g_i)|$$

If all $|G : C_G(g_i)|$ are divided by $p$, then $|Z(G)|$ must be divided by $p$, contradiction.
Therefore, for some $1 \leq i \leq r$, $p$ does not divide $|G : C_G(g_i)|$. Let $H = C_G(g_i)$.
By the Lagrange's theorem, $|H|$ divides $|G|$, so $|H| = p^\alpha \cdot k$ where $p^\alpha \nmid k$, $k \mid m$.
(If $|H|$ does not have factor of $p^\alpha$, then $p$ divides $|G : C_G(g_i)|$, which is contradiction.)
And, since $g_i \notin Z(G)$, $H < G$. Therefore, by the assumption of induction, $H$ has a sylow $p$-subgroup, so it is sylow $p$-subgroup of $G$.

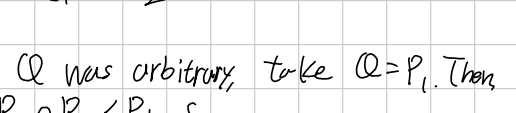
To show the second and third assertion, for group $G$ of order $p^\alpha m$, there exists a Sylow $p$-subgroup $P$ of $G$, by the first assertion. Let

$$S = \{gPg^{-1} \mid g \in G\} = \{P_1, P_2, \cdots, P_r\}$$

and let $Q$ be any $p$-subgroup of $G$.
Let $Q$ acts by conjugation on $S$. Write $S$ as a disjoint union of orbit:

$$S = \mathcal{O}_1 \cup \cdots \cup \mathcal{O}_s \quad \text{where } |\mathcal{O}_1| + \cdots + |\mathcal{O}_s| = r$$

(Note that $S$ has only one orbit under $G$, but it may have more than one under $Q$.)
Renumber the elements of $S$ so that the first $s$ numbers elements of $S$ are representatives of the $Q$-orbit $\mathcal{O}_i$: $P_i \in \mathcal{O}_i$.




It follows that $|\mathcal{O}_i| = |Q : N_Q(P_i)|$.
By definition, $N_Q(P_i) = N_G(P_i) \cap Q$ and
by the Lemma above, $= P_i \cap Q$.
Therefore, $|\mathcal{O}_i| = |Q : P_i \cap Q|$, $1 \le i \le s$. Since $Q$ was arbitrary, take $Q = P_1$. Then $|\mathcal{O}_1| = |Q : Q \cap Q| = 1$. Now, for $2 \le i \le s$, $P_1 \cap P_i < P_1$, so

$$|\mathcal{O}_i| = |P_1 : P_1 \cap P_i| > 1 \quad \text{for all } 2 \le i \le s.$$

Since $P_1$ has order of power of $p$, $|\mathcal{O}_i|$ must be power of $p$. Therefore,

$$r = \underbrace{|\mathcal{O}_1|}_{1} + \underbrace{|\mathcal{O}_2| + \cdots + |\mathcal{O}_s|}_{\text{order of } p} \equiv 1 \mod p. \quad \cdots (*)$$

Let $Q$ be any $p$-subgroup of $G$. Suppose that $Q$ is not contained in $P_i$, for all $i$.
Then, $Q \cap P_i < Q$ for all $1 \le i \le r$, therefore $|\mathcal{O}_i| = |Q : Q \cap P_i| > 1$ for each $1 \le i \le s$.
Since $Q$ has order of power of $p$, so $|\mathcal{O}_i| > 1$ is power of $p$,
therefore all $|\mathcal{O}_i|$ $(1 \le i \le s)$ are divided by $p$, it contradicts to $(*)$.
Consequently, $Q \le gPg^{-1}$ for some $g \in G$, which proves the second.
Let $Q$ be any Sylow $p$-subgroup of $G$, by the argument above, $Q \le gPg^{-1}$
for some $g \in G$, And, since $|Q| = |gPg^{-1}|$, $Q = gPg^{-1}$.
Therefore, all sylow $p$-subgroups are conjugate. Finally, for any $P \in Syl_p(G)$,

$$\#Syl_p(G) \overset{\text{def}}{=} n_p = |G : N_G(P)|, \text{ since } Syl_p(G) \text{ is the orbit.}$$

Moreover, $m = |G : P| = |G : N_G(P)| \cdot |N_G(P) : P| = n_p \cdot |N_G(P) : P|$, so $n_p \mid m$. ▨

[Corollary.
Let $P$ be a Sylow $p$-subgroup of $G$. Then the following are equivalent.
1). $P$ is the unique Sylow $p$-subgroup of $G$.
2). $P$ is normal in $G$
3). $P$ is characteristic in $G$.
Proof. Start 1). Since $Syl_p(G)=\{P\}$, the Sylow theorem gives that for all $g\in G$, $gPg^{-1}\in Syl_p(G)$, that is, $gPg^{-1}=P$. Therefore $P\trianglelefteq G$.
Conversely, given $Q\in Syl_p(G)$, $Q$ is conjugate of $P$, that is $Q=gPg^{-1}$ for some $g\in G$. Since $P\trianglelefteq G$, $Q=gPg^{-1}=P$.
And, since the automorphism preserves the order, therefore 1)=>3).
Finally, 3)=>2) trivially, by the inner automorphism, so 3)=>1).

Group of order of 15 is cyclic.
Let $G$ be a group of order 15.
By the Sylow theorem, there are only one Sylow 3-subgroup and Sylow 5-subgroup, denote $P$, $Q$ respectively. ($n_3 \equiv 1 \bmod 3$ and $n_3 \mid 5 \Rightarrow n_3 = 1$) ∎

Let $x, y \in G$ be generators of $P$, $Q$ respectively.
Since $P \trianglelefteq G$ is normal, $G = N_G(P)$
Meanwhile, Consider an action of $G$ on $P$ as

$$G \times P \to P : (g, h) \mapsto ghg^{-1}.$$

And the offered reprsentation $\varphi : G \to S_P$. Then, $\ker \varphi = C_G(P)$, therefore $G/\ker \varphi = G/C_G(P)$ is a subgroup of $Aut(P)$. But, since $P \cong \mathbb{Z}_3$, $Aut(P)$ has order of 2, therefore $G/C_G(P)$ must be trivial, that is, $C_G(P) = G$.
Now, $x$ and $y$ commute, so $|xy| = 15$, so $G = \langle xy \rangle$.

To show $G$ must be abelian, suppose that $Z(G) = \{1\}$.
By the Lagrange's theorem, every element of $G$ has order 3 or 5, except 1.
If all non-identity elements have order 3, then the class equation gives

$$15 = 1 + \sum |G : C_G(g_i)| = 1 + k \cdot 5 \text{ for some } k \in \mathbb{N},$$

because $C_G(g_i)$ must contains $\langle g_i \rangle \cong \mathbb{Z}_3$ and if $C_G(g_i) > \langle g_i \rangle$, then $C_G(g_i) = G$ by the Lagrange's theorem, it contradicts to $g_i \notin Z(G)$.
But, $15 - k5$ is divided by 5 but 1 is not divided,
therefore, $G$ must have order 5 element. Put this $x \in G$.

Group of order $pq$ ($p<q$ are prime, $p \nmid q-1$).
If $Z(G) \neq 1$, then $G/Z(G) \cong \mathbb{Z}_p$ or $\mathbb{Z}_q$ by the Lagrange Theorem, then $G$ must be abelian, since $G/Z(G)$ is cyclic.
Now, suppose that $Z(G) = 1$.
By the Lagrange Thm, every non-identity element of $G$ has order $p$ or $q$.
If there is no order $q$ element, then the class equation gives

$$pq = 1 + \sum [G : C_G(g_i)] = 1 + kq,$$

because $C_G(g_i) = \langle g_i \rangle \cong \mathbb{Z}_p$. (since $g_i \notin Z(G)$, $C_G(g_i) \neq \langle g_i \rangle$).
This is contradiction, since $q \nmid 1$. So, $G$ has a order $q$ element, denote this to $x$.
Let $H = \langle x \rangle$. Since $[G : H] = p$ is the smallest prime dividing $|G|$, so $H$ is normal.
Now, $G/H = N_G(H)/C_G(H)$ is a subgroup of order $p$, and isomorphic to a subgroup of $Aut(H)$.
Since $Aut(H) \cong \mathbb{Z}_q^\times$ has order of $q-1$, this implies $p \mid q-1$, contradiction.

「Group of order of $pq$, $p<q$. ($p, q$ are prime)
Let $P \in Syl_p(G)$ and $Q \in Syl_q(G)$.
Since $n_q$ divides $p$, $n_q = 1$ or $p$, but $n_q \equiv 1 \bmod q$, thus $n_q = 1$. $\therefore Q \trianglelefteq G$.
1. Case of $p \nmid q-1$.
By the Sylow Thm, $n_p = 1$ or $q$, but if $n_p = q$, then $q \equiv 1 \bmod p$, $p \mid q-1$, contradiction.
Therefore, $n_p = 1$, $P \trianglelefteq G$. Now, $G$ must be a cyclic, because;
Since $|P|$ and $|Q|$ are primes, both are cyclic. Let $P = \langle x \rangle$ and $Q = \langle y \rangle$.
Since $P \trianglelefteq G$, $N_G(P) = G$. Therefore,

$$G/C_G(P) = N_G(P)/C_G(P) \cong S \quad \text{for some } S \leq Aut(\mathbb{Z}_p)$$

But, $|Aut(\mathbb{Z}_p)| = p-1$, thus $S$ is trivial. (because $|S|$ divides $|G| = pq$).
Now, $G = C_G(P)$, and $x \in P$, $x$ and $Y$ are commute. This implies $|xy| = pq$. $G = \langle pq \rangle$.
2. Case of $p \mid q-1$
Let $Q$ be a Sylow $q$-subgroup of $S_q$, the symmetric group $q$ letters.
Since $q$ is prime, $|N_{S_q}(Q)| = q \cdot (q-1)$, because

1). Since $|S_q| = q!$, $|Q| = q$.
2). $S_q$ has $(q-1)!$ $q$-Cycles.
3). Since order of $gQg^{-1}$ is $q$, these are all cyclic.
4). $gQg^{-1} \cap Q = \{1\}$ for all $g \in G$, so

$$n_q = \#\{gQg^{-1} \mid g \in G\} = (q-1)!/(q-1) = (q-2)!.$$

Therefore, $n_q = G_Q = [S_q : N_G(Q)] = (q-2)!$, or $N_{S_q}(Q) = q(q-1)$. ]
$p$
Now, since $p \mid q-1$, $p$ divides order of $N_{S_q}(Q)$, so it has a subgroup of order $p$.
And, since $P \leq N_{S_q}(Q)$, $PQ$ is subgroup of order $pq$.
And, $C_{S_q}(Q) = Q$, $PQ$ is non-abelian group.

$$|G : N_{S_q}(Q)| = G_Q$$

Group of order 48 is Not simple.

$48 = 2^4 \cdot 3$. Therefore,

$n_3 = 1$ or $4$ or $16$ and $n_2 = 1$ or $3$.

If $n_2 = 1$, then $PG\,Syl_2(G)$ is the Normal, so there is nothing to prove.

Suppose $n_2 \neq 1$, or $n_2 = 3$. That is, there are three sylow 2-subgroup,

$$P_1, P_2, P_3 \subseteq G. \quad (\#P_i = 16).$$

For each $1 \leq i < j \leq 3$,

$$|P_i P_j| = \frac{|P_i| \cdot |P_j|}{|P_i \cap P_j|} = \frac{2^8}{|P_i \cap P_j|} \leq 48 = |G| \quad \text{if } k, P_i = P_j$$

So, $\frac{2^4}{3} \leq |P_i \cap P_j|$, $|P_i \cap P_j|$ cannot be $1, 2, 4, 16$ therefore

$$|P_i \cap P_j| = 8.$$

Take $Q = P_1 \cap P_2$, as subgroup of order 8. Since $Q \subseteq P_1, P_2$ and index 2, $Q \trianglelefteq P_1$ and $Q \trianglelefteq P_2$. That is, for any $g \in P_1 \cup P_2$,

$$gQg^{-1} = Q.$$

therefore, $P_1 \cup P_2 \subseteq N_G(Q)$. Since $|P_1 \cap P_2| = 8$,

$$|P_1 \cup P_2| = |P_1| + |P_2| - |P_1 \cap P_2| = 24$$

Now, if $|N_G(Q)| = 24$, then it has index 2 to $G$,

$$N_G(Q) \trianglelefteq G$$

And, if $|N_G(Q)| = 48$, then $N_G(Q) = G$, so $Q \trianglelefteq G$.

Group of order 30 is not simple.

$30 = 2 \cdot 3 \cdot 5$, let $P \in Syl_5(G)$ and $Q \in Syl_3(G)$.

If either $P$ or $Q$ is normal, then $PQ \leq G$, and it is group of order 15, because $|PQ| = \frac{|P| \cdot |Q|}{|P \cap Q|} = 15$. (All elements of $P$ has order 5 except identity, and " $Q$ " 3 " ).

So, $PQ$ is index of 2 to $G$, so $PQ$ is normal subgroup.

Now, assume neither $P$ nor $Q$ is normal. By the Sylow Theorem, $n_5 = 6$ and $n_3 = 10$. By the Lagrange's Theorem, for any $P_1, P_2 \in Syl_5(G)$, $P_1 \cap P_2 = \{1\}$. So, there are $6 \cdot 4 = 24$ distinct order 5 elements.

Similarly, there are 20 distinct order 3 elements, $24 + 20 = 44 \leq 30$, Contradiction. Therefore, either $P$ or $Q$ must be normal, so $PQ$ is index 15 subgroup.

In sum. Group of order 30 must have a subgroup of order of 15.

Group of order 105 is not simple.

$105 = 3 \cdot 5 \cdot 7$, so

$$n_3 = 1 \text{ or } 7 \qquad n_5 = 1 \text{ or } 21 \qquad n_7 = 1 \text{ or } 15$$

If it is simple, then $n_3 = 7$, $n_5 = 21$, $n_7 = 15$. Therefore there are

$$\underbrace{2 \cdot 7}_{\text{order 3}} + \underbrace{4 \cdot 21}_{\text{order 5}} + \underbrace{6 \cdot 15}_{\text{order 7}} = 14 + 84 + 90 \geq 130$$

So, contradiction by Counting

Group of order 12 is Not Simple.

Every group of order 12 has a normal sylow 3-subgroup or $G \cong A_4$.

$12 = 2^2 \cdot 3$. If $n_3 = 1$, then $G$ is not simple, it has a normal sylow 3-subgroup.

Suppose that $n_3 \neq 1$. Then, there is only possibility is $n_3 = 4$.

Let $P \in Syl_3(G)$. By the sylow Theorem, $|G : N_G(P)| = n_3 = 4$,

so $P = N_G(P)$, by $P \subseteq N_G(P)$ and $|N_G(P)| = 12/4 = 3$, given by the Lagrange Thm.

Now, consider an action of $G$ on $Syl_3(G)$ by conjugation:

$$G \times Syl_3(G) \to Syl_3(G) : (g, P) \mapsto gPg^{-1}$$

Since $Syl_3(G) = \{P_1, P_2, P_3, P_4\}$, we can write the permutation representation of this action:

$$\varphi : G \to S_4 : g \mapsto \sigma_g \quad \text{where} \quad \sigma_g : S_4 \to S_4 : P_i \mapsto gP_ig^{-1}$$

Now, the elements of $\ker \varphi$ normalizes all sylow 3-subgroups. That is,

$$\ker \varphi \leq N_G(P) = P$$

Since $P$ is not normal in $G$, $\ker \varphi = 1$. That is, $\varphi$ is injective,

$$G \cong \varphi[G] \leq S_4.$$

Since $G$ contains 8 elements of order 3, by Counting,

and $S_4$ has exactly 8 element of order 3, and all these are even,

Thus $G \cong A_4$.

「Groups of $p^2 \cdot q$, $p$ and $q$ distinct primes.

Let $G$ be a group of order $p^2 \cdot q$. Let $P \in Syl_p(G)$ and $Q \in Syl_p(G)$.

「In case of $p > q$, then $n_p = 1$ or $1 + kp$ $(k \geq 1)$, but $1 + kp \nmid q$, so $n_p = 1$.

Thus $P \trianglelefteq G$.

If $|G|=60$ and $G$ has more than one Sylow 5-subgroup, then $G$ is simple.

Prof. $60 = 5!$.. Since $n_5 > 1$. Then only possibility is $n_5 = 6$.

Assume there is non-trivial proper normal $H \subsetneq G$ exists.

If $5 \mid |H|$, then it has at least one sylow-5 subgroup, and since $H \trianglelefteq G$, all sylow 5-subgroups are contained in $H$. So,

$$|H| \geq 1 + 4 \cdot 6 = 25,$$

therefore $|H| = 30$. but, since order of 30 has a unique normal sylow 5-subgroup, so contradiction. Thus, $5 \nmid |H|$.

If $|H| = 6$ or $12$,